WP2.45

Tactile Localization in Forearm-Mounted Multi-Tactor Displays: The Effect of Tactor Number and Position

Dong-Geun Kim, Suheon Nam, Geunho Lee, Chaeyong Park and Seungmoon Choi

Interaction Laboratory, Pohang University of Science and Technology (POSTECH).

IEEE World Haptics Conference 2025

Research Goal

Evaluation of how different tactor positions and numbers on the forearm influence tactile localization accuracy.

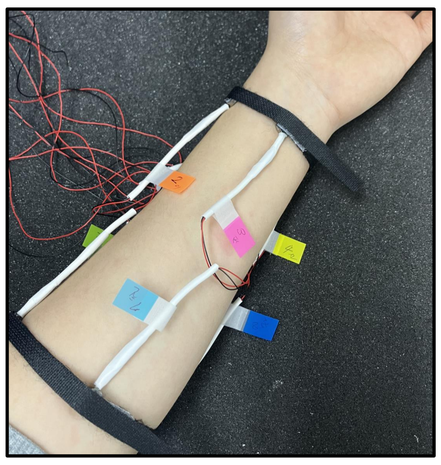
- We examined localization performance using **four, six, or eight tactors** mounted on the distal and/or proximal regions of the forearm.
- **The goal is to identify spatial configurations that support accurate tactile communication using wearable haptic interfaces.**

Methods

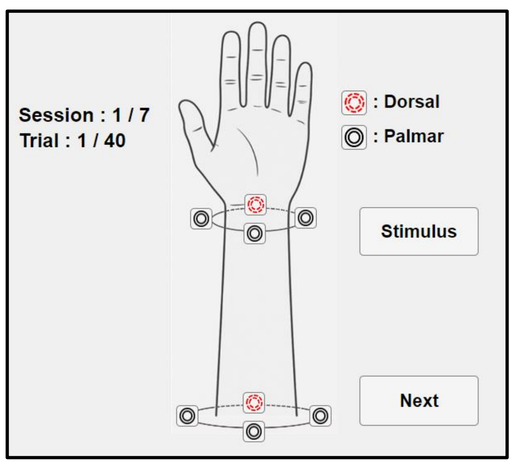
Participants & Setup.

- N = 10 (6M, 4F, age: 25.1 ± 2.4 years)
- ERM tactors (10 mm) driven by Arduino Due
- Mounted via Velcro bands on distal or proximal forearm
- Stimuli duration: 2 s, random order, 5 repetitions per tactor

(a)



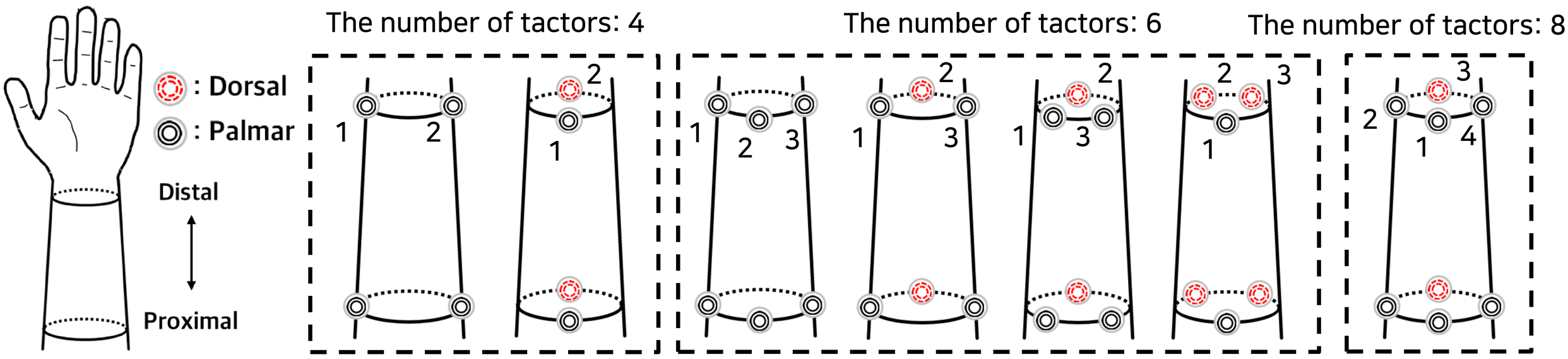
(b)



(a) Example setup showing placement of tactors.
(b) Graphical user interface (GUI) displaying the positions of tactors during the experiment.

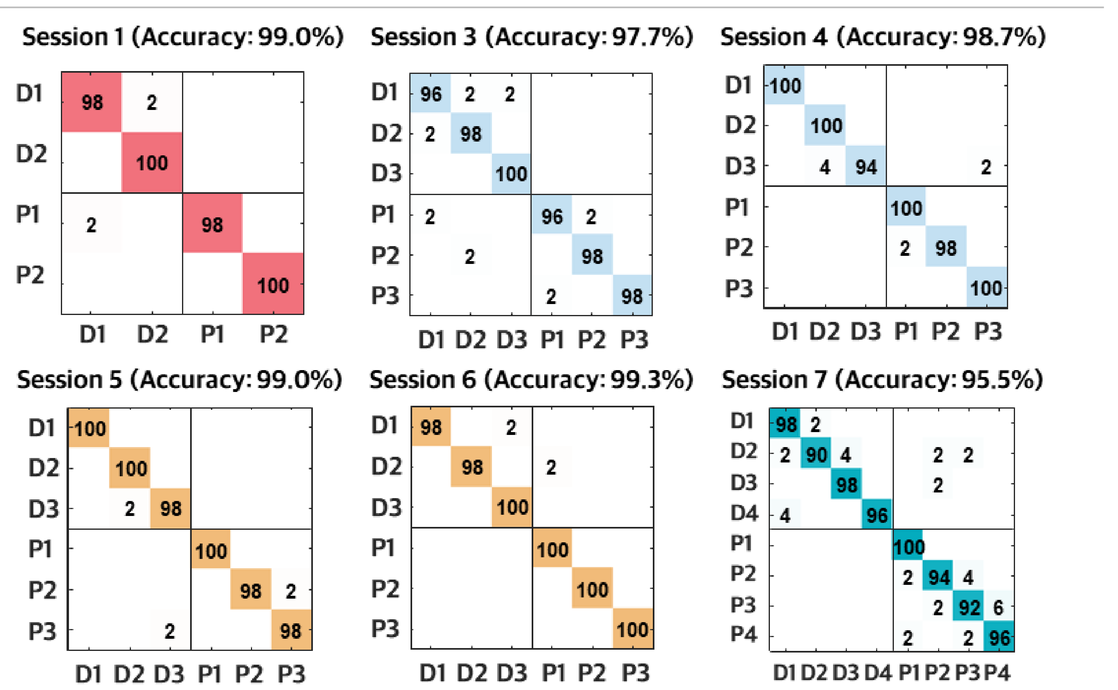
Configurations.

- 7 sessions with different layouts
- Configurations varied by anatomical side (palmar, dorsal, medial, lateral)



Results and Discussion

- **4 tactors (99%)**: **Near-perfect performance** regardless of configuration, indicating high spatial discriminability with sparse setups.
- **6 tactors (98%)**: **Slight drops** observed when tactors were placed on medial/lateral sides—possibly due to **skeletal propagation** reducing cue distinctiveness (needs further study).
- **8 tactors (96%)**: Denser layout may have caused **perceptual overlap,** making it harder to distinguish closely spaced stimuli.
- **Accurate localization is feasible with up to eight tactors**, though increased count may slightly degrade performance.

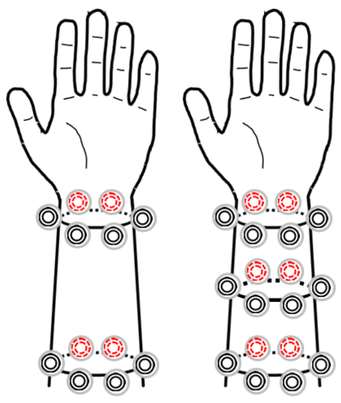


'D' = distal, 'P' = proximal; numbers indicate position index (e.g., D1, P1).

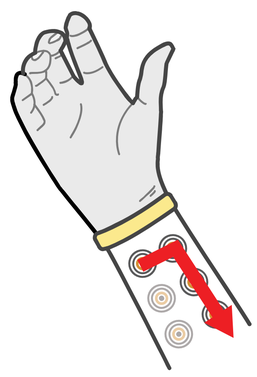
Future Work

- (a) We aim to explore **denser spatial layouts** by using more than eight tactors and expanding the coverage area across the entire forearm.
- (b) We will also investigate how **temporal parameters (e.g. duration)** affect localization to optimize **spatio-temporal tactile encoding**.
- (c) Ultimately, our goal is to design **tactile languages** that enable communication through wearable vibrotactile displays.

(a)



(b)



(c)